

Resistance $\rightarrow$ property of material that restricts the movement of electrons through it.
(ohm)

Resistor $\rightarrow$

is a two-terminal passive electrical element that provides electrical resistance to current flow.
used to limit and regulate current flow.

$\rightarrow$ only receive energy

ohm's law $\rightarrow$

voltage across a conductor is directly proportional to the electric current flowing through the conductor provided that all physical conditions such as area, length, temperature, etc. remain constant.

$$V \propto I$$

$$V = RI$$

$\hookrightarrow$ resistance.

BJT (Bipolar junction transistor) $\rightarrow$

3 terminal semiconductor device composed of combinations of p-type and n-type semiconductors.

consists of pn junction.

magnify / amplify the signal.

Emitter

• heavily doped.

Base

• slightly doped.

collector

• moderate

EB junction

CB junction

Active region

$\rightarrow$ transistor $\rightarrow$ on

[amplifier]

CB $\rightarrow$ reverse, EB $\rightarrow$ forward.

$$I_c = \beta I_B$$

saturation :- ON.

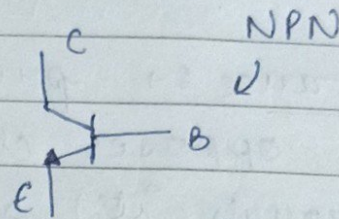
CB, EB $\rightarrow$ forward.

$$I_c = I_{sat}$$

cut-off region :- CB, EB $\rightarrow$ reverse.

OFF.

$$I_B = I_C = 0.$$



PNP
 $\downarrow$
base

NPN
 $\downarrow$
base.

current is due to holes.

(-||-)
Capacitor :-

used to store electric charge

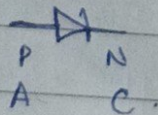
measure of capacitor's ability to store charge.
capacitance.

Farad
(F)

$$C = Q/V$$

ability of capacitor to store electric charge.
per unit voltage.

Diode :-



conducts electricity only in 1 direction.

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Inductor :-

2 terminal passive electrical element that stores energy in the form of magnetic field when electric current flows through it.

SI unit $\rightarrow$ Henry (H)

Inductance :- property of a conductor that opposes change in electric current flowing through it by generating a magnetic field.

Laws

Faradays

1st law :- changing magnetic field induces an electromotive force.

2nd law :- mag. of induced emf is proportional to rate of change of magnetic flux through the coil.

LED (light emitting diode) :- A tiny bulb that glows when electricity flows through it one direction.